

MIRZA MD. SHAFI UDDIN

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Location: Dhaka 1212, Bangladesh

PROFESSIONAL SUMMARY

AI and Data Science enthusiast with hands-on experience in Large Language Models (LLMs), AI agent development, and full-stack development. Skilled in building AI-driven solutions, deploying SaaS products, and developing scalable applications with a strong foundation in machine learning and API development. Currently specializing in AI orchestration and model evaluation through the Ostad AI Engineering Bootcamp.

CORE SKILLS

- **AI & Engineering:** LLM Integration (OpenAI, Gemini APIs) , AI Agents , Prompt Engineering , RAG Pipelines, SLM Benchmarking, Vector Databases (Redis).
- **AI Infrastructure:** Retrieval-Augmented Generation (RAG), Hybrid Search (BM25 + Vector), Cross-Encoder Reranking, Vector DBs (Pinecone, ChromaDB, Redis).
- **ML & Data:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, SQL.
- **Backend & MLOps:** FastAPI (Asynchronous), Node.js, Docker, CI/CD Gated Eval Pipelines, PostgreSQL, MongoDB, DigitalOcean, Hostinger, Git, Ragas and TruLens.
- **Frontend:** React.js, Tailwind CSS, Responsive UI Design
- **Language:** Python, JavaScript, Java.

WORK EXPERIENCE

AI Agent & LLM Intern – Mevrik (Dec 2025 – Feb 2026)

- **Autonomous Agents:** Engineered and deployed AI agents to automate CX workflows, implementing stateful logic and tool-calling to reduce manual intervention.
- **LLM Optimization:** Refined prompt architectures and conversational memory, increasing model accuracy and response grounding by measurable margins.
- **SaaS Integration:** Integrated Gemini capabilities into a production SaaS environment using FastAPI and Webhooks for real-time AI processing.

Junior Engagement Officer – X Integrated Marketing Agency (June 2024 – October 2024)

- Assisted in planning and executing digital marketing campaigns to increase audience engagement. Coordinated with creative and technical teams to ensure smooth campaign delivery and improved customer interactions.

COURSES & CERTIFICATIONS

AI Engineering Bootcamp | Ostad | In Progress

- Specializing in building production-ready AI systems with an emphasis on Agentic AI and RAG reliability.
- Implementing Hybrid Search, Cross-Encoder Reranking, and hallucination detection using Ragas and TruLens. Developing and fine-tuning Small Language Models (SLMs) for domain-specific tasks.

LEADERSHIP EXPERIENCE

Secretary (IT Department) | Robotics Club of BRAC University | Feb 2024 – June 2025

- Managed IT operations and provided technical support for workshops, events, and robotics competitions.
- Led digital initiatives and contributed to robotics projects, including an Arduino-based soccer robot.

EDUCATION

- **B.Sc. in Computer Science | BRAC University (June 2021 – October 2025) | CGPA: 3.22 / 4.00**
- **Higher Secondary Certificate (HSC) | Ispahani Public School and College (2019) | GPA: 4.67 / 5.00**
- **Secondary School Certificate (SSC) | Rampur High School (2017) | GPA: 4.91 / 5.00**

PROJECTS

Thesis: The Exiled Prince – A Human-Centered Digital Game for Promoting Cybersecurity

[This Report](#)

Awareness and Engagement among Non-CS University Students in Bangladesh

- Designed and developed a gamified digital experience aimed at enhancing cybersecurity awareness and engagement among university students.
- Implemented story-driven mechanics and branching choices to boost engagement, learning effectiveness, and user motivation based on HCI principles.

Production RAG System with Hybrid Retrieval

[Project Link](#)

- Developed a domain-specific "Ask My Docs" system featuring BM25 + Vector search and Cross-Encoder Reranking for high-precision document retrieval.
- Implemented citation enforcement to eliminate hallucinations, ensuring all AI responses are verifiable.
- Stack: Python, LangChain, Pinecone, FastAPI, Docker.

Local SLM Benchmarking & Deployment Suite

[Project Link](#)

- Deployed and benchmarked Llama 3, Mistral, and Phi-3 locally using Ollama to evaluate privacy-first AI alternatives.
- Analyzed hardware-specific trade-offs between quantization levels, inference latency, and memory throughput.
- Stack: Python, Ollama, Streamlit, Linux/Bash.

The Role of Campus Living Distance in Shaping Academic Outcomes

[Project Link](#)

- Conducted data-driven analysis to examine how commuting distance influences academic performance, using real-world datasets and statistical modeling.
- Visualized findings and recommended insights for improving student engagement and institutional planning.
- **Tools:** Python, Pandas, NumPy, Matplotlib/Seaborn, Jupyter Notebook

Hospital Management Portal – Full-Stack Web Application for Healthcare

[Project link](#)

- Developed a secure hospital portal enabling appointment scheduling, user management, and role-based access control for staff and patients.
- Implemented role-based authentication, dashboard analytics, and integrated chatbot support for efficient clinical workflow and user engagement.
- **Tools:** MERN Stack, JWT, Socket.io, Bootstrap, Postman

ACHIEVEMENTS

- 3rd Position Soccer Bot Competition TechSpectra (Robotics Club of BRAC University)
- 5th Position Soccer Bot Competition (IUBAT)

REFERENCE

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